

Summary of Selected Blockchain-Based Federated Learning Papers

1. A Trusted Federated Learning Method Based on Consortium Blockchain

- Proposes a system called **CBFL**.
- Combines federated learning with a consortium blockchain.
- Uses two blockchains:
 - One chain for global model aggregation.
 - Separate chains for local client training.
- Groups clients based on their computing power to avoid slow training.
- Uses Proof of Authority and smart contracts.
- Tested on FashionMNIST and CIFAR-10.
- Shows better accuracy and faster training than FedAvg, especially with non-IID data.

2. Blockchain-Based Privacy-Preserving Federated Learning for IoV

- Proposes a method called **BPFL**.
- Focuses on secure federated learning for vehicles.
- Encrypts model updates using homomorphic encryption.
- Uses an improved Multi-Krum method to remove malicious updates.
- Blockchain manages trust and rewards.
- Tested on MNIST and BelgiumTS datasets.
- Maintains good accuracy while resisting poisoning attacks.

3. Distributed Intrusion Detection for Vehicular Networks

- Proposes a framework called **NNFLB-NET**.
- Uses federated learning to train intrusion detection models.
- Blockchain is used to store model updates and ensure trust.
- Tests different neural networks such as LSTM and GRU.
- Uses the VeReMi dataset.
- Achieves up to 94% detection accuracy.

4. Blockchain for Improving Federated Learning Integrity

- Presents a conceptual framework (no full implementation).
- Uses blockchain as a trust layer for federated learning.
- Stores model updates on the blockchain.
- Uses IPFS to store large model files.
- Suitable for sensitive domains like healthcare and finance.
- Improves transparency but adds system overhead.

5. Blockchain and Federated Learning for Industrial IoT (Survey)

- Survey paper focused on Industrial IoT security.
- Reviews how blockchain and federated learning work together.
- Federated learning protects industrial data privacy.
- Blockchain ensures secure and trusted model sharing.
- Discusses open challenges and future directions.

6. Blockchain and Federated Learning for Healthcare IoT

- Proposes a framework called **FBCI-SHS**.
- Targets healthcare and medical IoT devices.
- Uses federated learning to train models on patient data.
- Blockchain controls data access and trust.
- Achieves high privacy and intrusion detection performance.

7. Blockchain-Based Federated Learning for Digital Advertising

- Proposes **FRL-BSC**.
- Uses federated reinforcement learning.
- Blockchain ensures transparent payment and fairness.
- Edge devices train models locally.
- Reduces fraud and system latency.

8. Blockchain Federated Learning for IoT (Survey)

- Survey of blockchain-based federated learning systems.
- Covers IoT domains such as vehicles, healthcare, and industry.
- Compares different system designs.
- Identifies blockchain overhead as a major issue.

9. PrivChain-AI for Financial Applications

- Proposes a system for secure financial data analysis.
- Combines federated learning with multiple privacy techniques.
- Uses blockchain to manage trust.
- Tested on fraud detection datasets.
- Achieves high prediction accuracy.

10. Robust Federated Learning with Blockchain

- Proposes **FedGenBlk**.
- Uses genetic algorithms to improve training.
- Blockchain verifies model updates.
- Works well with non-IID data.
- Resistant to data poisoning attacks.

11. Blockchain-Based Federated Learning: Survey

- Broad survey covering blockchain-based federated learning.
- Reviews systems, applications, and challenges.
- Highlights communication and computation overhead.
- Suggests need for lightweight blockchain designs.